A Dog's Dream - Project 8 UML Class Diagram

ng game framework

Project8_driver

+ main()
+ Ex1()

creates

CanvasFrame_P8

- frame : JFrame
- canvas : CanvasPanel_P8

+ CanvasFrame_P8()

contains

CanvasPanel_P8

- dog : Dog
- obstacles : List<Obstacle>
- clouds : List<Cloud>
- frameNumber : int
- score : int
- difficultyLevel : int
- scrollSpeed : static int
- spawnRate : int

+ CanvasPanel_P8()
+ Simulate()
+ updateDifficulty()
+ spawnObstacle()
+ paintComponent(Graphics)
+ drawGameOver(Graphics2D)
+ resetGame()
+ getScrollSpeed() : int

inner class

myActionListener

inner class of CanvasPanel_P8

+ keyPressed(KeyEvent)
+ keyReleased(KeyEvent)

has Dog

GameObject_P8 (abstract)

x : int
y : int
width : int
height : int

+ update() : abstract
+ render(Graphics) : abstract
+ overlaps(GameObject_P8) : boolean

Bird

- birdImage : BufferedImage

+ Bird(int,int,int,int)
+ render(Graphics)

Dog

- ySpeed : int
- dogImage : BufferedImage

+ Dog(int, int)
+ moveUp()
+ moveDown()
+ stop()

Cloud

- speed : int

+ Cloud(int, int, int)
+ update()
+ render(Graphics)

Obstacle (abstract)

speed : int
ySpeed : int

+ Obstacle(int,int,int,int,int,int)
+ update()
+ isOffScreen() : boolean
+ render(Graphics) : abstract

FlyingSquirrel

- squirrelImage : BufferedImage

+ FlyingSquirrel(int,int,int,int)
+ render(Graphics)

TennisBall

- tennisBallImage : BufferedImage

+ TennisBall(int,int,int,int)
+ render(Graphics)

Legend: + public - private # protected

Hollow triangle arrows show inheritance. Labeled arrows show creates/contains/has relationships.